Morning Paper Report - Monday, June 01, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, SEM/intelligence/creativity, and egocentric predictive-coding memory themes.

1. Working memory and processing speed as predictors of general fluid intelligence in adults with high functioning autism: A structural equation modeling approach.

M. Alhwaiti
Applied neuropsychology. Adult, 2026

Source: Semantic Scholar; matched terms: working memory, modeling, intelligence, structural equation

Link: https://www.semanticscholar.org/paper/090520d96948885c47333c231d6856b276a543b1

Goal: This study aimed to evaluate the predictive power of working memory and processing speed on general fluid intelligence (Gf) in adults with high functioning autism

Method: One-hundred-seven participants were included in the study

Results: This shows that 64.1% of the total variance in general Gf of those who participated in the study is accounted for by the combination of working memory capacity and processing speed.

Conclusion: The two independent variables (working memory capacity, processing speed) when put together yielded a coefficient of multiple regression (R) of 0.667 and a multiple correlation square of 0.641 This shows that 64.1% of the total variance in general Gf of those who participated in the study is accounted for by the combination of working memory capacity and processing speed.

2. Redundancy Repartitions Capacity in Visual Working Memory: The Sample-Size Model of Neural Activation

Authors unavailable
2026

Source: OSF/PsyArXiv; matched terms: visual working memory, working memory, precision

Link: https://osf.io/preprints/psyarxiv/8x4wg_v1/

Goal: Visual working memory (VWM) has a limited capacity of around four items, yet recall is also improved by structure in the memory array

Method: We reconcile these views by testing how color redundancy — a low-level structural regularity — alters recall precision and response time in a continuous color-reproduction task using a between-subjects sample (21 participants) and a small-N within-subjects design (5 participants)

Results: Visual working memory (VWM) has a limited capacity of around four items, yet recall is also improved by structure in the memory array

Conclusion: Redundant and non-redundant item precision improved in one participant, suggesting a shared capacity improvement — a ‘spill-over’ benefit

3. Associations of Prenatal Socioeconomic Status and Childhood Working Memory: A Structural Equation Modeling Approach

Shelley H Liu, David C. Bellinger, K. Dams-O’Connor, Jeanne A. Teresi, I. Pantic, S. Martínez-Medina, et al.
Children, 2025

Source: Semantic Scholar; matched terms: working memory, modeling, structural equation

Link: https://www.semanticscholar.org/paper/3ea74b5520c0858236213cc8a2bba9045cfd9895

Goal: Objective: To determine if prenatal socioeconomic status (SES) is associated with childhood working memory (WM), we constructed a more precise, integrative measure of WM using variables from multiple tasks that may provide a more representative measure of WM

Method: Prenatal SES was measured using the Mexican Association of Marketing Research and Public Opinion Agencies (AMAI) index

Results: Results: Children had a mean age of 6.6 years [SD 0.6], and 50.5% were boys

Conclusion: Higher prenatal SES was significantly associated with higher childhood WM in females (standardized factor loading = 0.26; p = 0.002), but not in males Conclusions: Prenatal socioeconomic status is a predictor of childhood working memory, but it may be a stronger predictor for girls compared with for boys.

4. Medial temporal lobe lesions reduce visual working memory precision.

Choo Y, Thakurdesai SP, Qadri A, Fruchet OE, Jackson SN, Chatterjee R, et al.
Brain : a journal of neurology, 2026

Source: PubMed; matched terms: visual working memory, working memory, precision

Link: https://pubmed.ncbi.nlm.nih.gov/41108071/

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

5. Object Details Persist in Visual Working Memory Regardless of Category Sufficiency for Search

Authors unavailable
2026

Source: OSF/PsyArXiv; matched terms: visual working memory, working memory, attentional guidance

Link: https://osf.io/preprints/psyarxiv/cfr7e_v1/

Goal: When searching for objects in complex environments, performance often improves with more specific target representations

Method: Across two experiments, observers searched consecutively for images of real-world objects from the same basic-level category

Results: We found that while fewer object-specific details were maintained in VWM when category membership was sufficient, the amount of detail only diminished across repetitions of the exact same object

Conclusion: We therefore examined whether the format of target representations used for search is sensitive to the sustained preservation of object-category membership